Watch The Build Video

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Introduction

Sheet good prices are out of control these days. This miter saw station takes the unique approach of using readily available dimensional lumber to build an equally useful workspace. With careful planning and the accounting for of wood movement, dimensional lumber can save you tons of money when building out your shop. Tired of the cookie cutter miter saw stations that flood the Youtube algorithm? Then follow along with these plans to create a miter saw station that is creatively designed and crafted.

Notes

- ❏ Read the entire plan before starting the project.
- ❏ As with most woodworking, dimensions may be slightly off depending on the tolerances and accuracy of your build. Variances in material thickness can also play a factor. Use relative dimensions when appropriate.
- ❏ The plans were designed around dimensional lumber, but a hardwood species of your preference can be substituted.
- ❏ The use of a planer in this build is not necessary but recommended when applicable. Just note that this will change some of the dimensions based on the stock you have available to you.
- ❏ This build is designed with an overall dimension of 26 ⅛” deep x 92 ½” wide x 36” tall. Adjust these dimensions before starting the build to fit your space if needed.
- ❏ Some aspects of this build may vary between the plans and the build video. These plans are meant for the masses, so the sliding rail system is not outlined here. Watch and follow along with the build video if you’re interesting in imitating that function.
- ❏ A cutlist is not available for this as dimensional lumber lengths and widths may vary depending upon your location. All parts are listed with a recommended dimensional lumber type. This build is capable of being made entirely from 1x4, 1x6, and 2x4 material.

Tools and Supplies

Titebond III Wood Glue	https://geni.us/4JCcR
Ridge Carbide Dado Stack	https://lddy.no/1g6ux
Oil Based Polyurethane	https://geni.us/tEfnI
Magswitch 150	https://magswitch.com/collections/woodworking/products/magswitch-magjig-150-8110005?ref=4msdrHDDEtPa1K
1500mm Linear Guide Bearings and Rail Pair	https://geni.us/qy8TIgH
22" Full Extension Drawer Slides 10 pack:	https://geni.us/A71QAF5
16" Full Extension Soft Close Drawer Slide	https://geni.us/1jatb
Drawer Pulls, 6" 10 pack	https://geni.us/IeGYjH
T-Track	https://geni.us/YRKPVo
Figure 8 Fasteners	https://geni.us/APB2opt
Scribe	https://geni.us/H93A5
Jig Saw	https://geni.us/i7HRN
Router Bit Corner Bead	https://geni.us/o3n1Y

Parts List

PART	Description	QTY	Dimensions	Material
A	Cabinet Side Vertical Frame	8	1 ½" x 3 ½" x 31 ¼"	2x4 Dimensional Lumber
B	Cabinet Side Horizontal Frame	8	¾" x 3 ½" x 17"	1x4 Dimensional Lumber
C	Cabinet Side Panel	16	¾" x 4 ¾" x 31 ¼"	1x6 Dimensional Lumber
D	Front Cabinet Stretchers	4	1 ½" x 3" x 27"	2x4 Dimensional Lumber
E	Rear Cabinet Stretchers	4	¾" x 3 ½" x 27"	1x4 Dimensional Lumber
F	Cabinet Toe Kicks	2	¾" x 5 ¼" x 27"	1x6 Dimensional Lumber
G	Bottom 2 Drawers Sides	8	¾" x 5 ½" x 22"	1x6 Dimensional Lumber
H	Bottom 2 Drawers Fronts/Backs	8	¾" x 5 ½" x 25 ¼"	1x6 Dimensional Lumber
I	Top 2 Drawers Sides	8	¾" x 3 ½" x 22"	1x4 Dimensional Lumber
J	Top 2 Drawers Fronts/Backs	8	¾" x 3 ½" x 25 ¼"	1x4 Dimensional Lumber
K	Drawer Bottoms	8	¾" x 21 ¼" x 25 ¼"*	1x6 Dimensional Lumber
L	Cabinet Countertop	1	¾" x 26" x 92 ½"**	1x6 Dimensional Lumber
M	Cabinet Countertop Lip	1	¾" x 1 ½" x 92 ½"	1x4 Dimensional Lumber
N	Upper Box Side	4	¾" x 3 ⅝" x 18 ¾"	1x6 Dimensional Lumber
O	Upper Box Top Panel	2	¾" x 30" x 18 ¾"***	1x6 Dimensional Lumber

*Overall dimensions of the drawer bottom listed. This piece can be made with plywood or by gluing 1x6 dimensional lumber into a panel.
** Cabinet top dimensions may need to be altered based on your space. Use 1x6" dimensional lumber glued up into a panel or MDF/Plywood can be substituted
*** Upper box top panel can be made from MDF/Plywood or by gluing up 1 x dimensional lumber ot create a panel.

Parts List

PART	Description	QTY	Dimensions	Material
P	Cabinet Drawer Sides	16	¾" x 22" x 5 ½" (8) ¾" x 22" x 3 ½" (8)	1x6 Dimensional Lumber & 1x4 Dimensional Lumber
Q	Cabinet Drawer Front/Back	16	¾" x 25 ¼" x 5 ½" (8) ¾" x 25 ¼" x 3 ½" (8)	1x6 Dimensional Lumber & 1x4 Dimensional Lumber
R	Cabinet Drawer Bottom Panel	8	¾" x 25 ¼" x 21 ¼"****	1x6 Dimensional Lumber
S	Upper Box Drawer Sides	4	¾" x 2 ¾" x 16 ¾"	1x4 Dimensional Lumber
T	Upper Box Drawer Front/Back	4	¾" x 2 ¾" x 26 ¾"	1x4 Dimensional Lumber
U	Upper Box Drawer Bottom Panel	2	¾" x 16" x 26 ¾"*****	1x6 Dimensional Lumber
V	Cabinet False Drawer Front Styles	16	¾" x 2 ¼" x 7 ⅝" (4) ¾" x 2 ¼" x 6 ⅝" (4) ¾" x 2 ¼" x 5 ⅝" (4) ¾" x 2 ¼" x 4 ⅝" (4)	1x4 Dimensional Lumber
W	Cabinet False Drawer Front Rails	16	¾" x 23" x 2 ¼" (4) ¾" x 23" x 2" (4) ¾" x 23" x 1 ¾" (4) ¾" x 23" x 1 ½" (4)	1x4 Dimensional Lumber
X	Cabinet False Drawer Front Panels	8	¾" x 23" x 3 ⅝" (2) ¾" x 23" x 3 ⅛" (2) ¾" x 23" x 2 ⅝" (2) ¾" x 23" x 2 ⅛" (2)	1x6 Dimensional Lumber & 1x4 Dimensional Lumber
Y	Upper Box False Drawer Front Styles	4	¾" x 2 ¼" x 3 ¼"	1x4 Dimensional Lumber
Z	Upper Box False Drawer Front Rails	4	¾" x 1" x 24 ½"	1x6 Dimensional Lumber
AA	Upper Box False Drawer Front Panels	2	¾" x 1 ¾" x 24 ½"	1x4 Dimensional Lumber

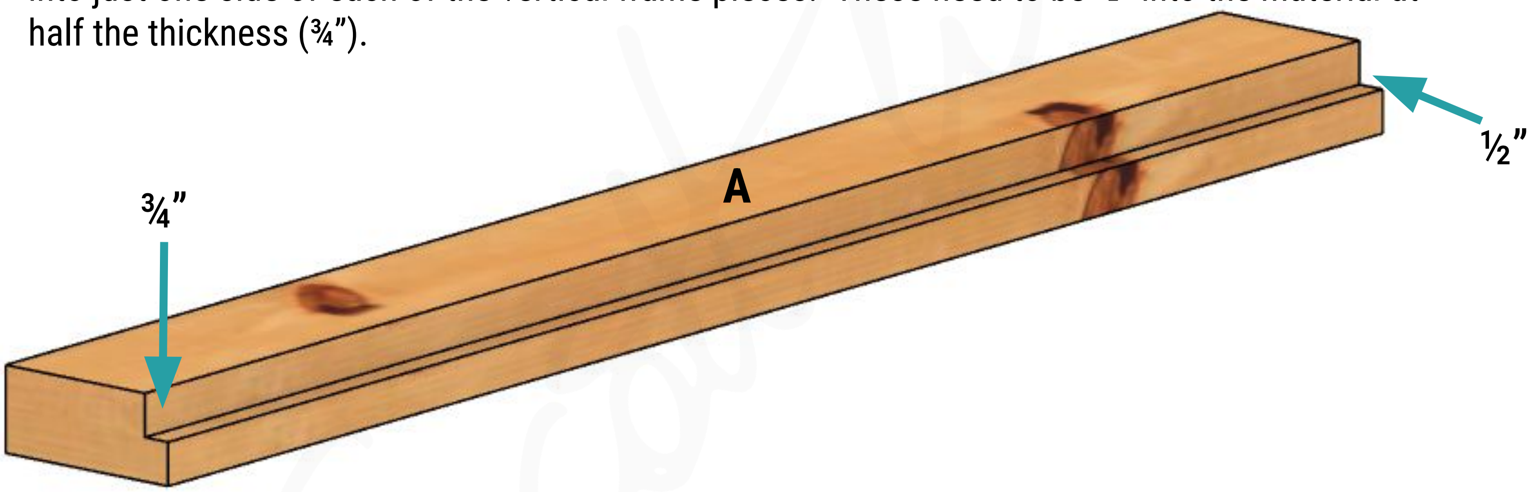
**** The cabinet drawer bottom panel can be created using 1 x material glued into a panel or MDF/Plywood can be substituted.
***** The upper box drawer bottom panel can be created using 1 x material glued into a panel or MDF/Plywood can be substituted.

Step 1

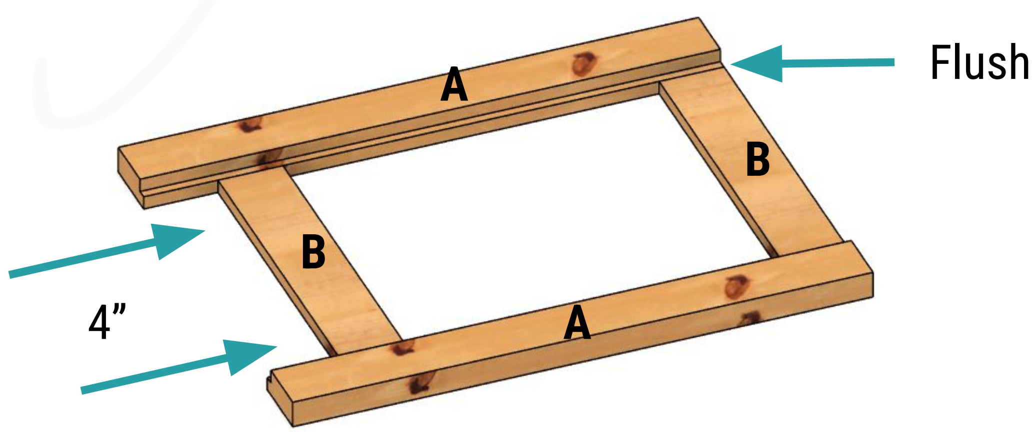
Cut & Assemble the Cabinet Side Frames

Each side frame of the cabinets is identical to the rest. This will help reduce the required tool set ups by building the sides at the same time. Note the overall cabinet frame size is 24” deep x 30” wide x 31 ¼” tall. The depth and height will be set in this step, so make any alterations to fit your needs at this time.

Start by cutting the frame pieces (Parts A & B) to size. These dimensions are listed in the parts list. The cabinet side vertical frame (Part A) need rabbets before assembly. These should be cut into just one side of each of the vertical frame pieces. These need to be ½” into the material at half the thickness (¾”).



Using ⅜” x 2” dowels for joinery (pocket holes or other methods can be substituted), glue and clamp the 4 cabinet frame sides together. It’s recommended to use 3 dowels at each joint. Be sure that the cabinet side horizontal frame (Part B) is flush with the outsides. Or in other words, opposite the rabbets that were cut in the cabinet side vertical frames (Part A).



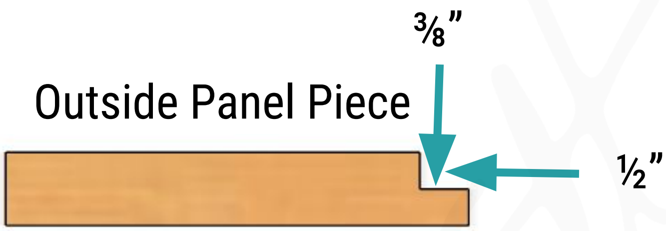
Step 2

Cut & Assemble the Cabinet Side Panels

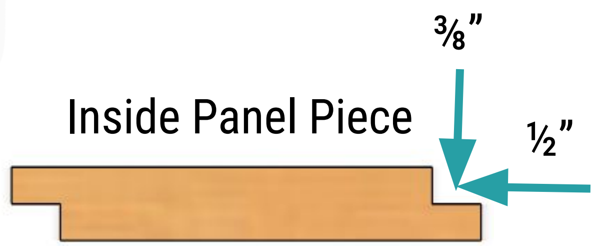
Cut the cabinet side panels (Part C) to their dimensions listed in the parts list. These panels will be constructed using a shiplap technique. The two outside panel pieces get only one rabbet cut into them while the two inner panel pieces get a rabbet cut on both sides as shown below. Be sure the inner panel pieces have rabbets opposite of each other to create a “Z” shape.

Top View Of Panel

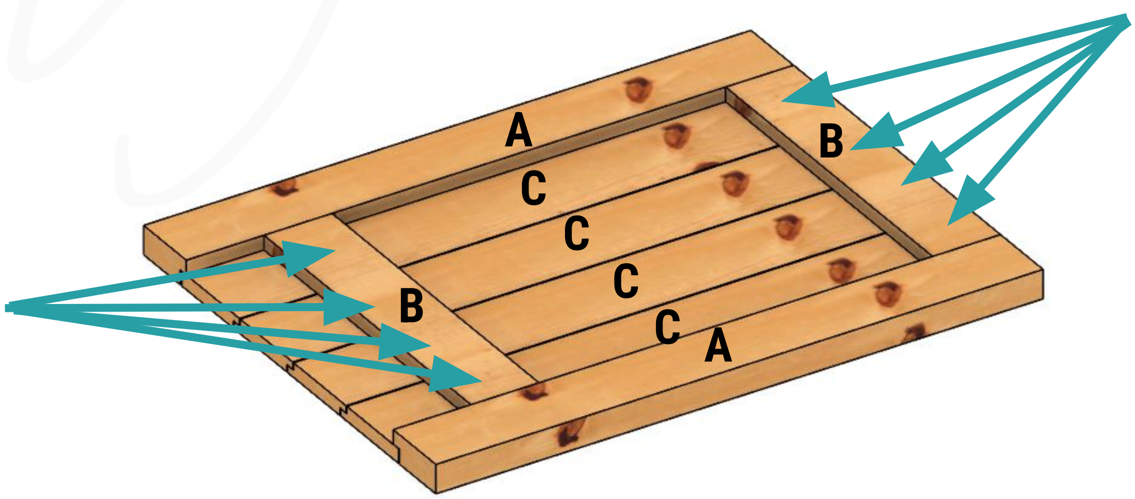
Outside Panel Piece



Inside Panel Piece



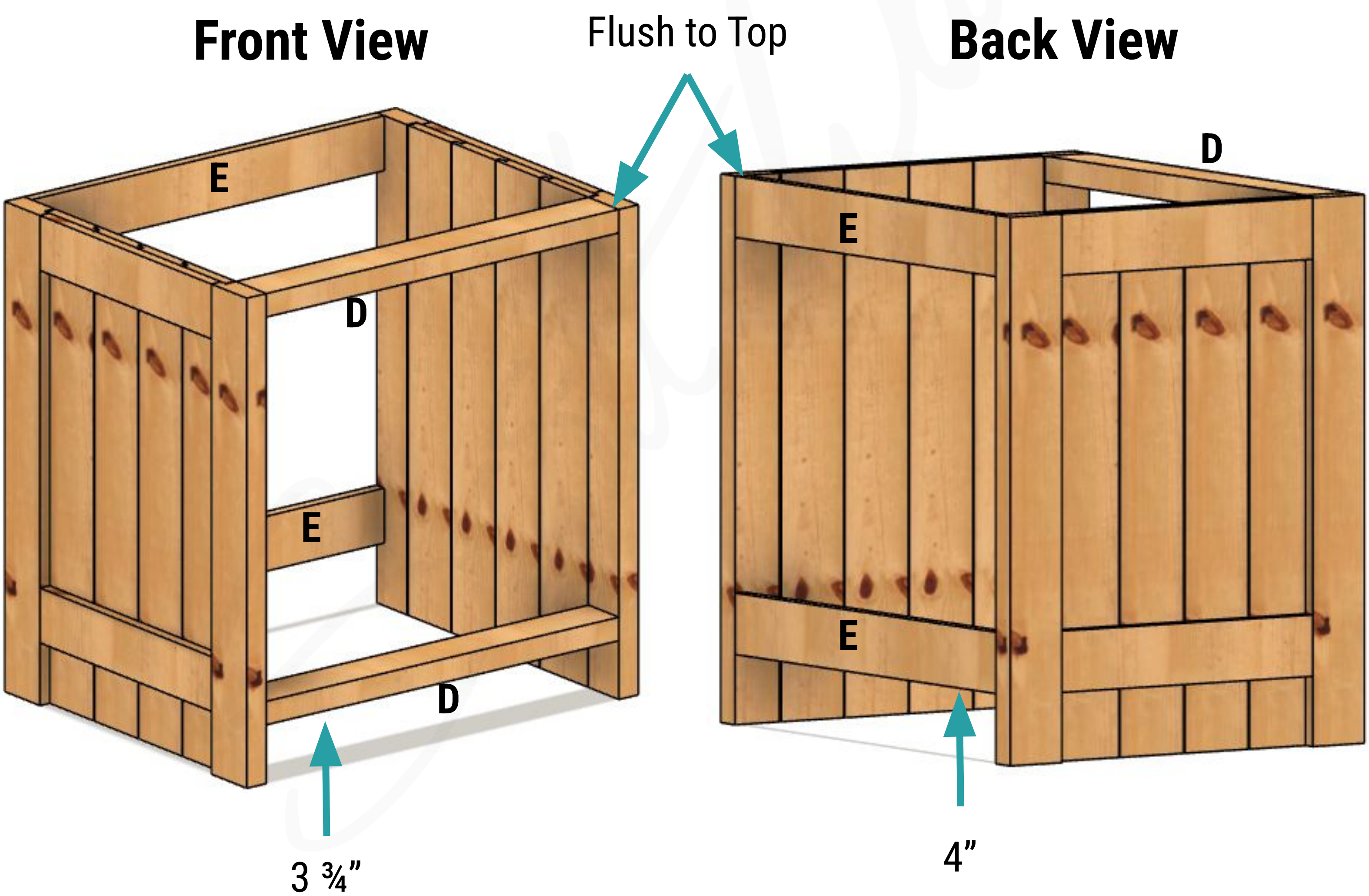
With the rabbets cut, place these on the assembled cabinet side frame from the previous step. Using a nail or screw in the center of each board, attach each panel to the frame. Nailing from the center of each board will allow for wood movement. Be sure to allow roughly 1/16” gap in the shiplap to allow for wood movement. These should be nailed or screwed from the inside to hide the fasteners.



Step 3

Join the Side Frames Together

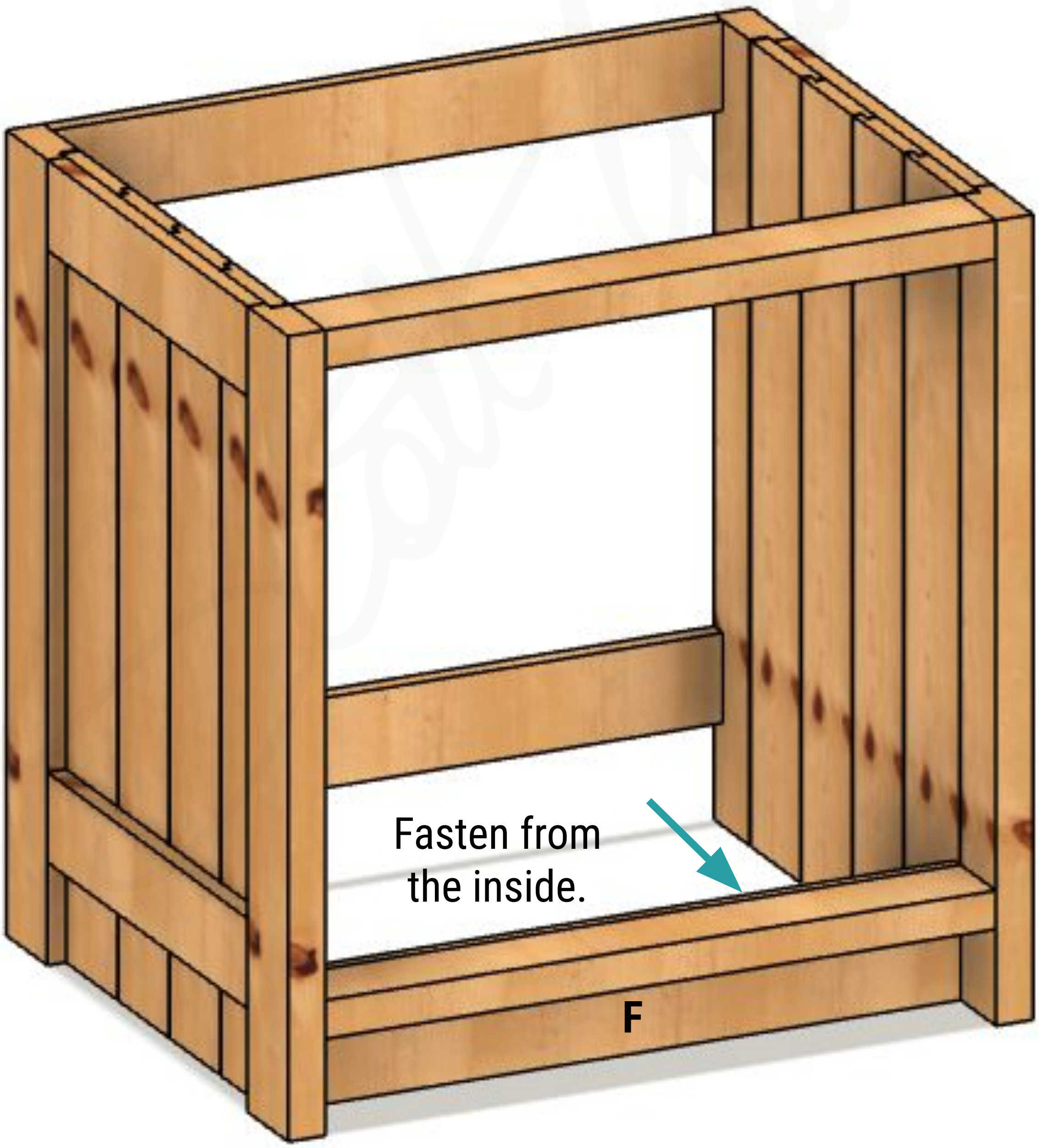
Cut the cabinet stretchers (Parts D & E) to the dimensions laid out in the parts list. Using the same dowel joinery method from Step 2, join the two sides of each cabinet assembly together.



Step 4

Add the Toe Kicks

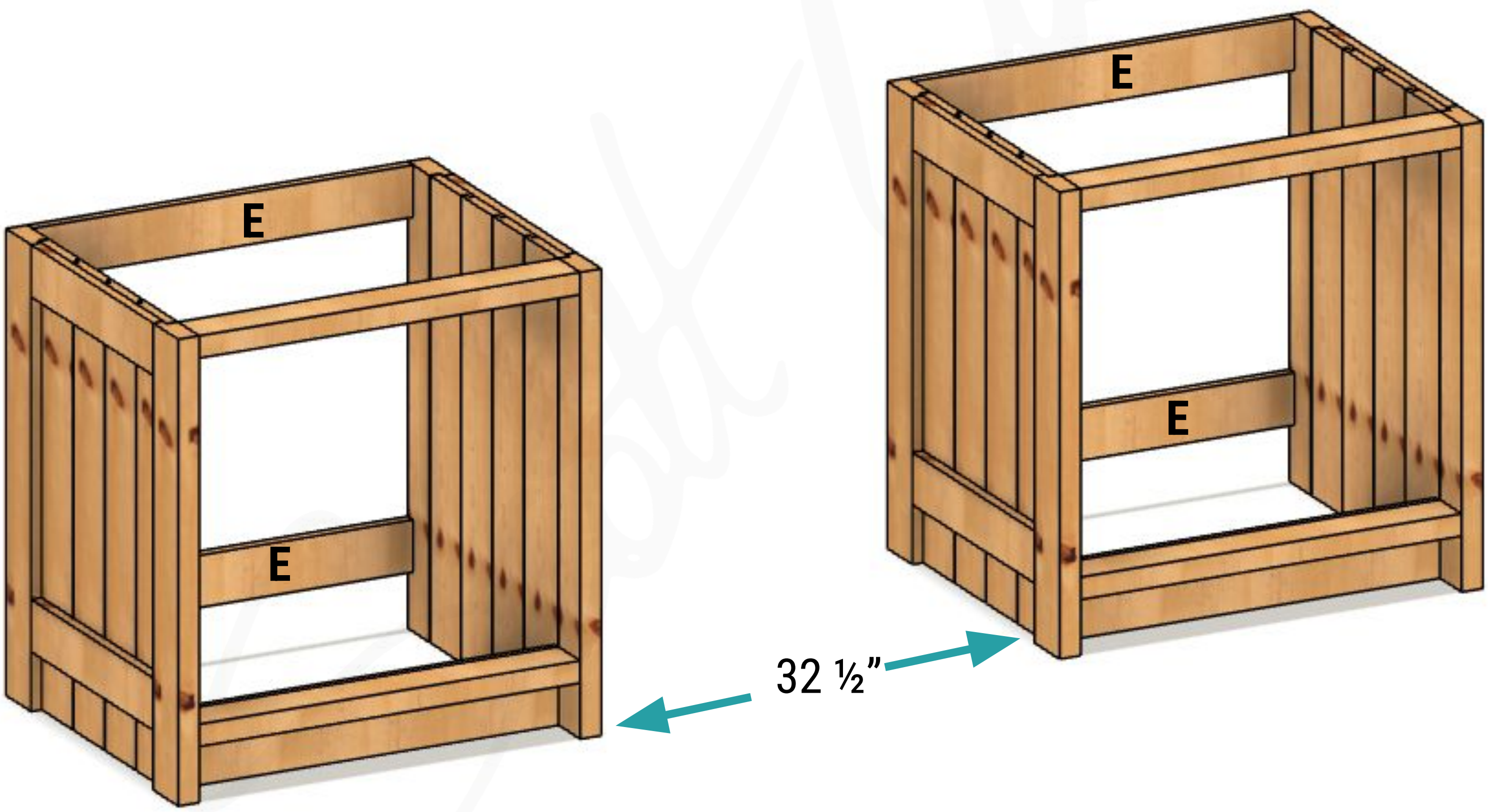
Cut the cabinet toe kicks (Part F) to size using the dimensions from the part list. From the inside of the cabinet, screw or nail the toe kicks to the bottom front stretchers.



Step 5

Attach the Cabinets to the Wall

There are many ways to fasten the cabinets to the wall. These include shims, leveling feet, or scribing the cabinets to your floor. Choose the best method for your situation. The cabinets should be secured into the wall using screws through the rear cabinet stretchers (Part E).

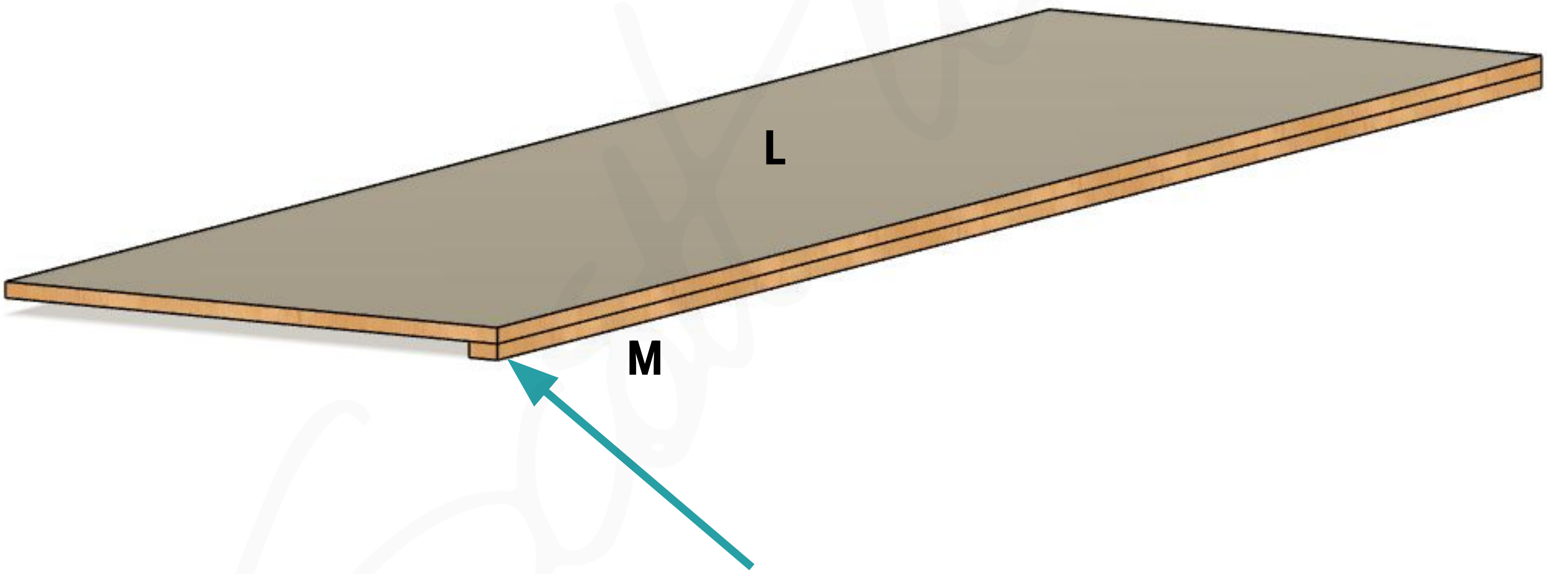


Be sure to measure the distance between the cabinets before securing them to the wall. For this example, the dimension between the cabinets is 32 ½”. This will affect the dimension of the cabinet top. So be sure to adjust this width to your needs or space.

Step 6

Assemble / Glue the Counter Top

If using only dimensional lumber, glue up a panel for the dimensions necessary for your cabinet countertop (Part L). You may choose to substitute MDF or plywood to create this part. The countertop should be flush with the sides of the outside of each cabinet. The front of the cabinet countertop will overhang the cabinets by 2 ⅛" in this design, but you may choose a dimension that's visually appealing to you. Depending on the orientation of the miter saw station within your own space, you may choose to do create the same overhang on one side or both. Just be sure to account for that extra distance when cutting the top to dimension.



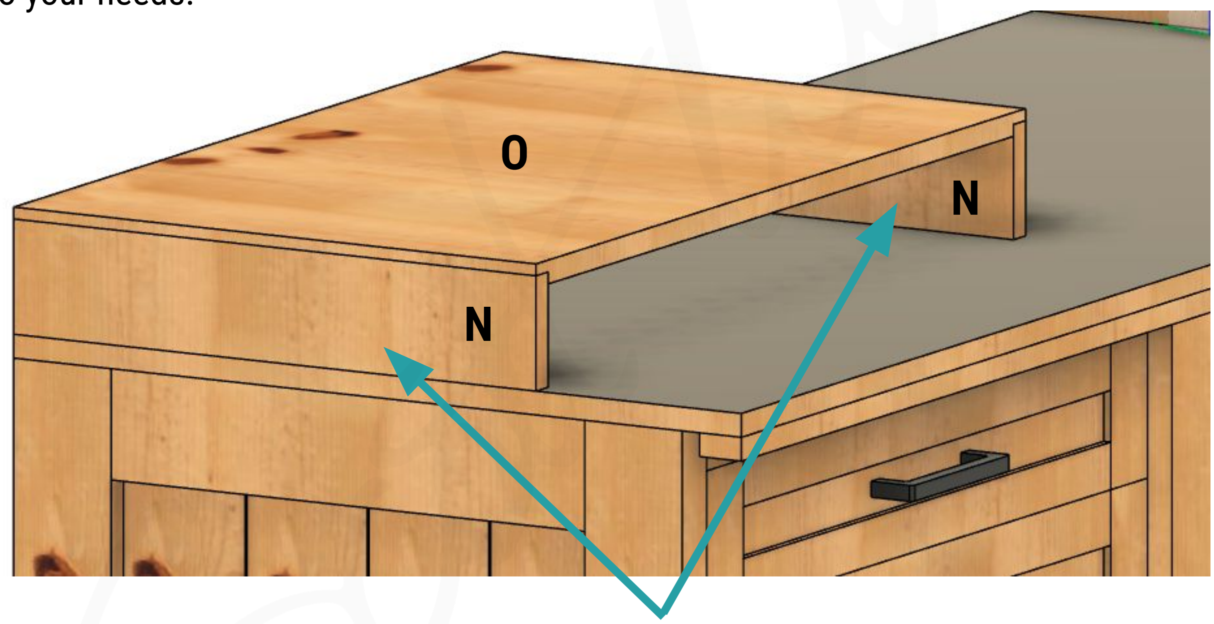
Attach the cabinet countertop lip (Part M) to the front of the cabinet counter top (Part L). This is a visual preference as well. It is not a functional piece and can be omitted based on your preferences. If you opted to overhang the top on either side, create a lip piece for the sides as well.

If applying a laminate top, take this time to do so.

Step 7

Create the Upper Boxes

The plans will have an upper box on each side of the miter saw. However, you may decide to create the sliding concept from the youtube video and construct only one box. Dimensions may vary depending on your exact miter saw. Just plan for the boxes to be slightly higher than your miter saw's cutting platform, as you can always shim to match the height. The depth of the upper boxes should compliment your exact saw as well. Below we will use 18 ¾" as the depth, but his can be adjusted to your needs.



The height of the upper box sides (Part N) should be your desired overall height minus ⅜" to account for the rabbeted upper box top panel (Part O).

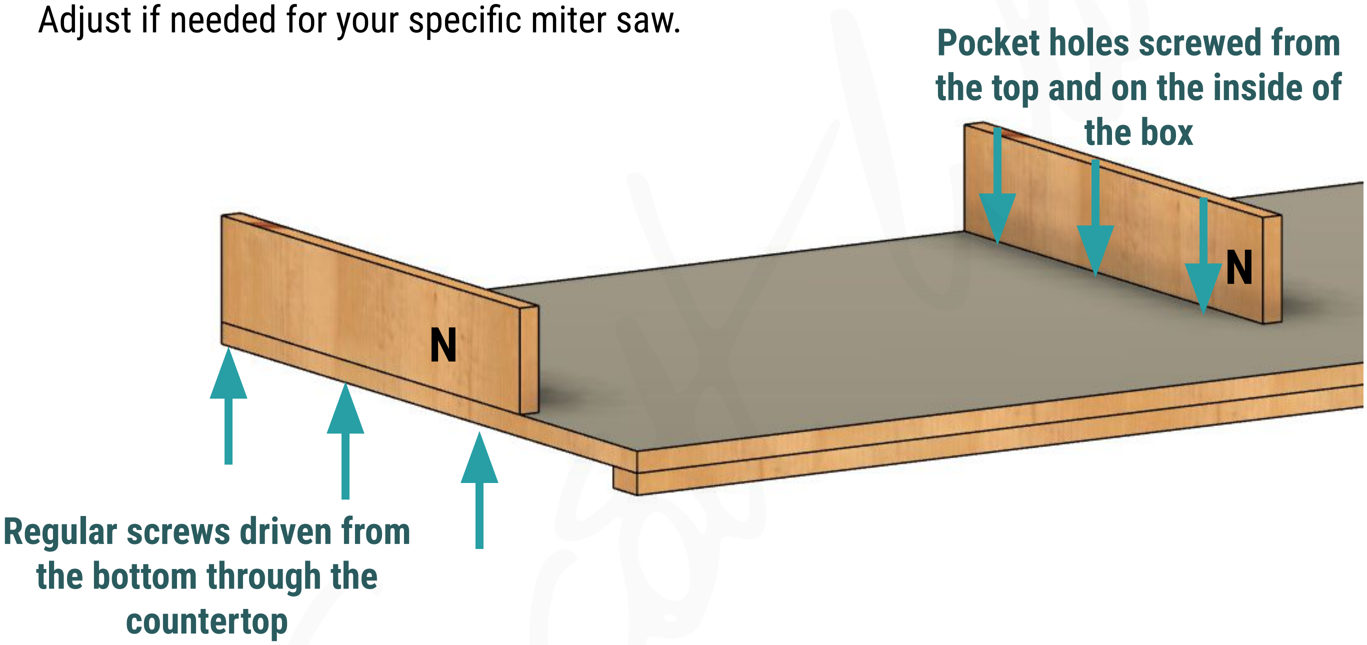
Cut the upper box sides (Part N) and the upper box top panels (Part O) to size according to the parts list, but adjusted for your specific miter saw's dimensions.

Create a ¾" rabbet on both sides of your upper box top panel (Part O) half the thickness of the panel (⅜").

Step 8

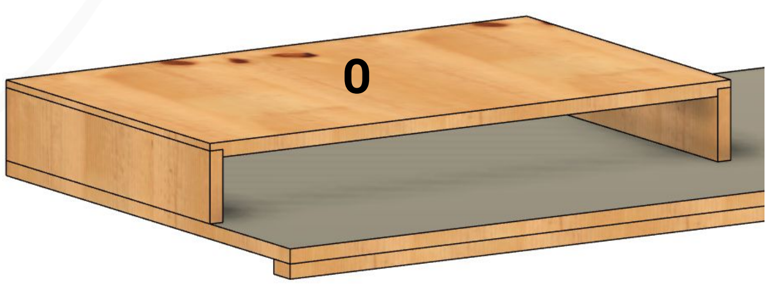
Install the Upper Boxes

Since the countertop has not be installed yet, you have the option of using pocket screws or screws driven from the bottom of the countertop to install the upper box sides (Part N). If using pocket screws, just be sure to orient them to the inside of the box so they will not be visible. These should be spaced out 28 ½" from each other if using the dimensions from the parts list. Adjust if needed for your specific miter saw.



Before the upper box top panel (Part O) can be glued on, you will want to install drawer slides. It will be difficult to access this space after. Just be sure to account for the drawer fronts created in a later step.

Glue the rabbeted upper box top panel (Part O) to the upper box sides (Part N)



You may choose to install the upper box drawers before glueing the upper box top panel (Part O). See Step 13

Step 9

Install T Track (Optional)

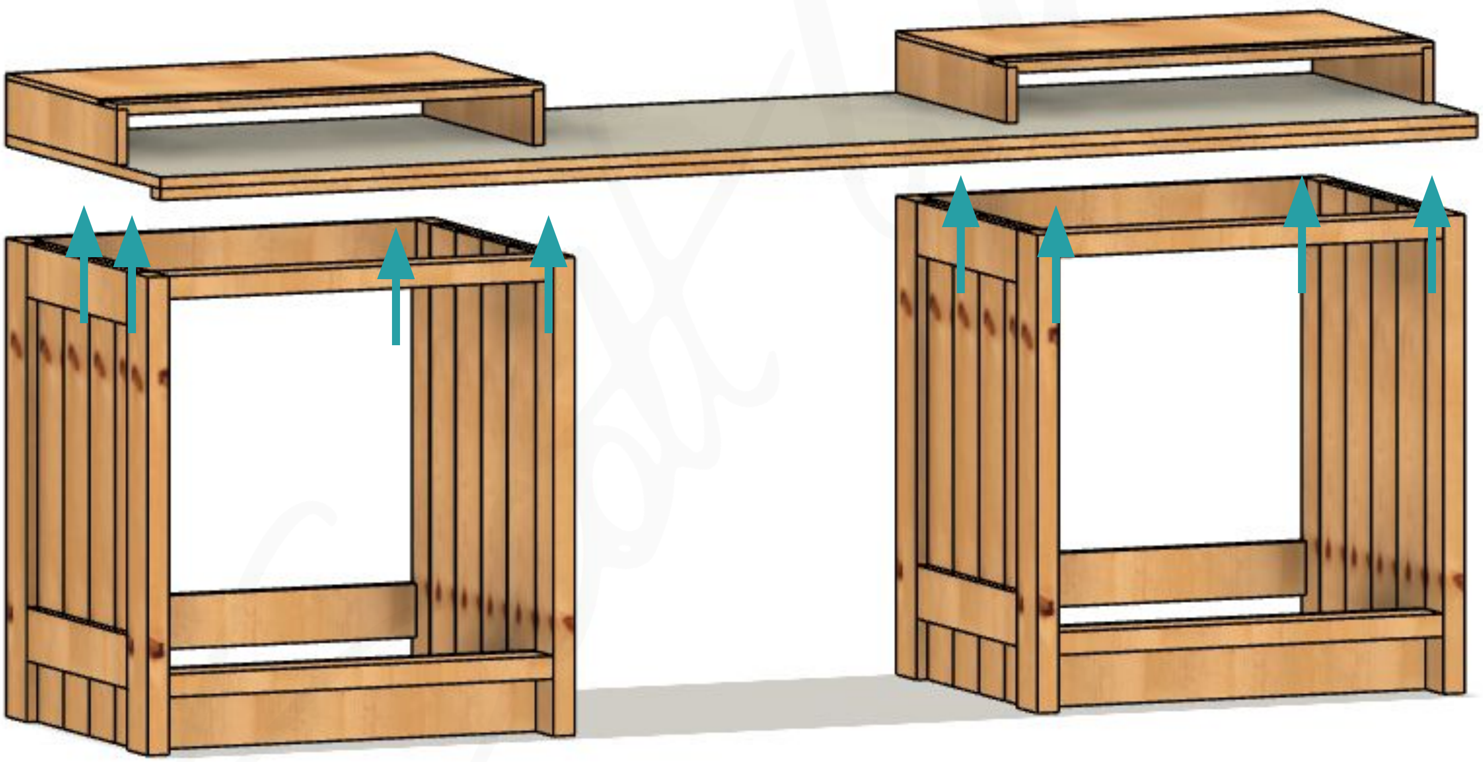
If planning to use T tracks in the top of the upper boxes, take this time to route out the required channels. Decide whether you want the T tracks in front of the miter saw's fence or behind it. Measure this distance for your specific miter saw. For this example, the T track was set 2 ¼" from the front and is intended to be used in front of the miter saw's fence.

Using CA Glue and screws, install the T track. Be sure the screws you use will not protrude through the bottom of the panel.

Step 10

Install the Counter Top

With the upper boxes installed, you can now install the the counter top. Just be sure to account for wood movement. Figure 8 fasteners are recommended. Place the figure 8 fasteners around the perimeter of your cabinet boxes.



From underneath using the figure 8 fasteners, drive screws into the countertop to create a secure attachment.

Step 11

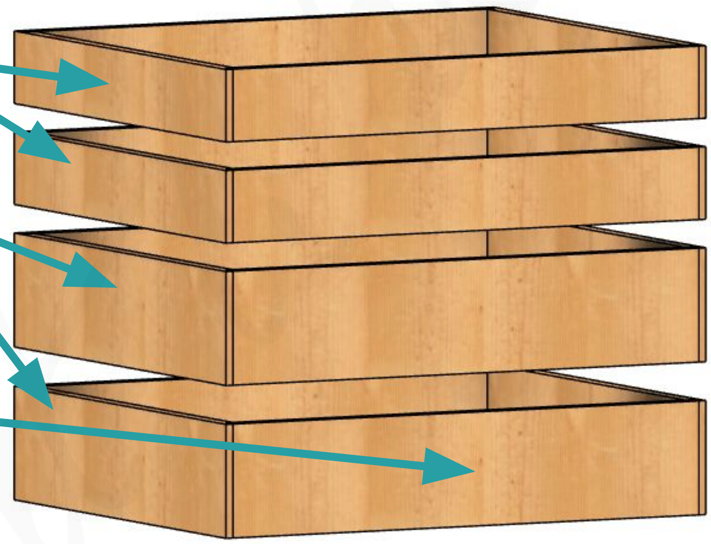
Create the Cabinet Drawer Boxes

For this build, 4 drawers were used, but you can choose to alter the plans to fit your storage needs. For simplicity, the top 2 and bottom 2 drawers are the same dimensions.

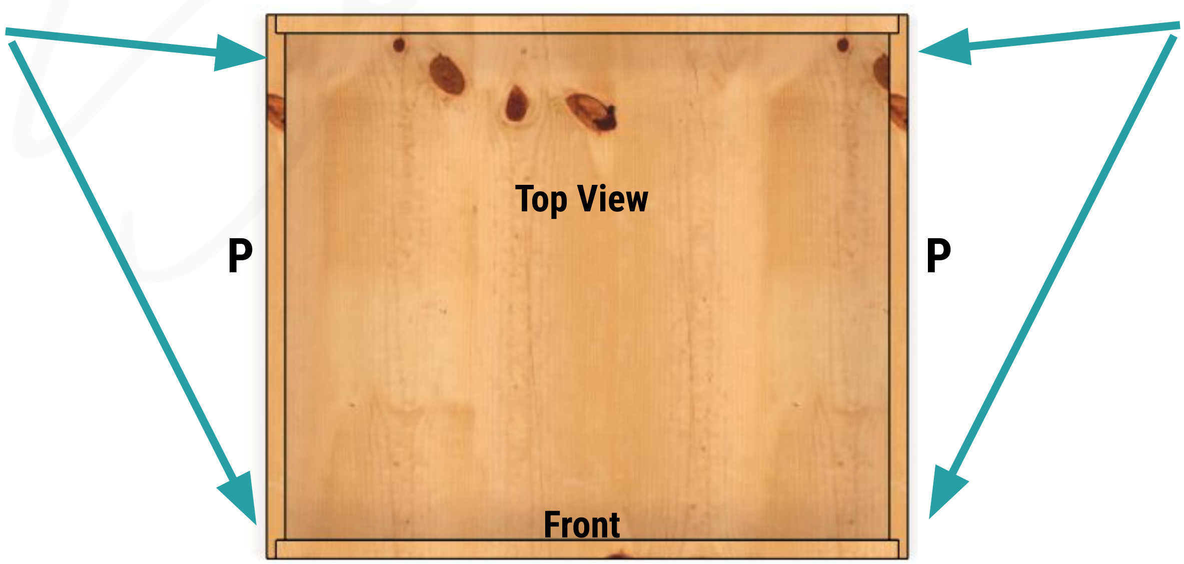
The height of the upper drawers is 3 ½"

The height of the lower drawers is 5 ½"

The width of the drawer should be width of the cabinet opening minus 1" to account for drawer slides, or in this case 26". That makes the front and back pieces 25 ¼" when accounting for the rabbeted joints



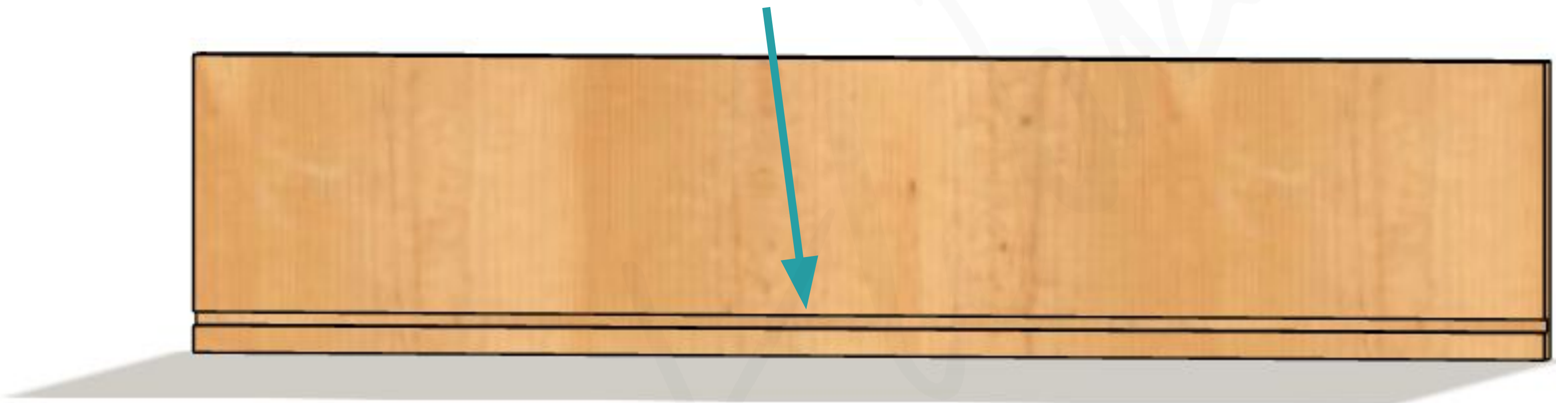
The drawers will be constructed using rabbets on the cabinet drawer sides (Part P). These will be done at ⅜" depth or half the thickness of the material. The front and back of the drawers will sit within the rabbets. Cut these rabbets now.



Step 12

Create the Cabinet Drawer Boxes Continued

All 4 walls of the drawer boxes will need a ¼" deep and wide dado cut ½" from the bottom to accommodate the cabinet drawer bottom panel (Part R).



A ⅜" wide by ½" deep rabbet must be cut from all 4 sides of the cabinet drawer bottom panel (Part R).

Glue the drawer boxes together. The rabbeted cabinet drawer bottoms should fit within the groove of the drawer walls. The image below on the left depicts this process with a wall removed for viewing. The image on the right is the final drawer assembly.

Step 13

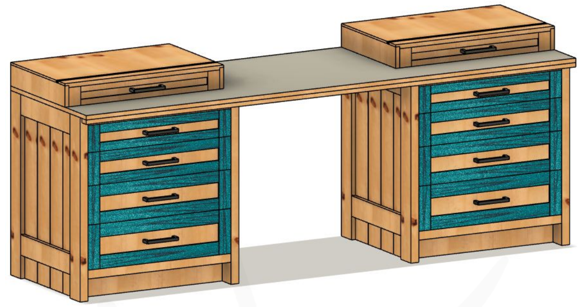
Create the Upper Box Drawers

The upper box drawers will be created in the same fashion as the cabinet drawer boxes. They will utilize the rabbeted side construction as well as the rabbeted drawer bottoms. Repeat the same exact steps as you did with the cabinet drawers. The only difference is the height of these drawer boxes, which will be 2 ¾".

There isn't much room to install these once the upper box top panels (Part O) are installed. You can choose to assemble these before gluing the panels on top.

With all of the drawer boxes completed. You can install them. Be sure to account for the thickness of the false drawer fronts since these are inset drawers. Refer to the build video for any additional help with installation of the drawers.

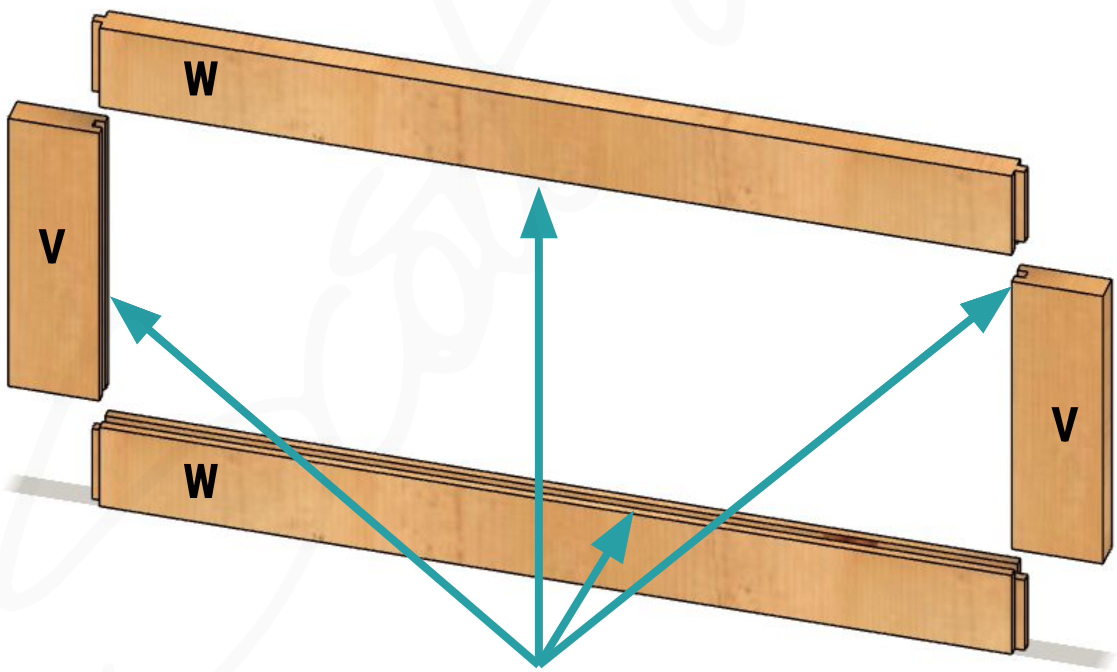
Step 14



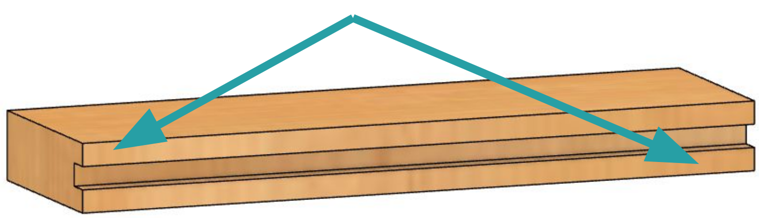
Create the Cabinet False Drawer Fronts

Frame and panel style drawer fronts were used in this project. But you can simply opt for solid piece drawer fronts. The measurements in the parts list are representative of the entire width of the opening. Adjust these to create your desired reveals. For example, if you're looking for 1/16" reveals, shorten the overall width of the drawers by ⅛" (2 x 1/16"). For the same example, the overall height of each drawer front should be reduced by 1/16", except the very top and bottom drawer which would be reduced by 3/32".

Accounting for the desired reveals, cut the rail and styles (parts V & W) of all the drawer fronts first. Note that the styles will all have the same widths, but the rails will vary in width to account for the size of the overall drawer fronts.



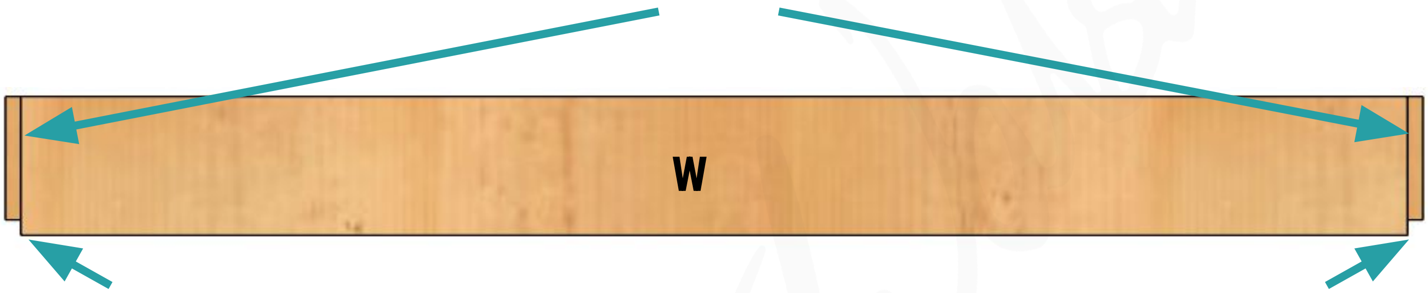
A dado should be cut on the inside of all rail and style pieces. It will be ¼" wide and ¼" deep. This will house the panel later. Take your time to dial in the set up so that there is ¼" material on each side of the dado.



Step 15

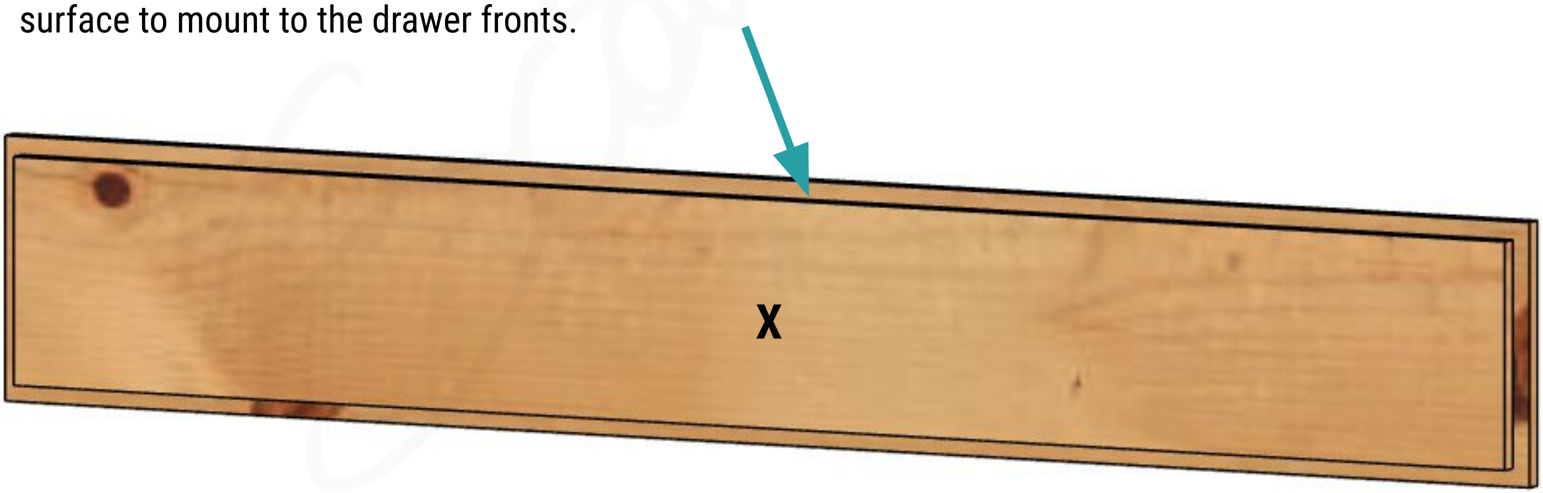
Create the Cabinet False Drawer Fronts Continued

Create the tenon on each end of the rails so they fit within the dado cut on the styles.



Note that the inside corners of each tenon need to be removed to allow for the panel to reside within the dados. Remove the ¼" piece necessary to do so.

The cabinet false drawer front panels (Part X) need to have a ¼" rabbet cut on all 4 sides. This should be ¼" deep as well. This allows the panel to sit within the grooves of the rails and styles. It also allows the back of the panel to be flush with the back of the rails and styles, allowing for a flat surface to mount to the drawer fronts.

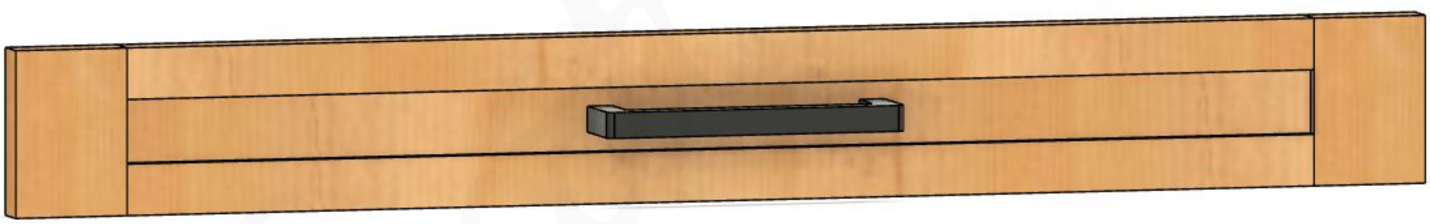


Step 16

Create the Upper Box False Drawer Fronts

The upper box false drawer fronts are created in the exact same fashion as the cabinet false drawer fronts. Just be sure to account for your desired reveals as done before with the cabinet false drawer fronts.

Mark for and install your desired drawer pulls.



Remove the drawer pulls for now and use the screw hole locations to temporarily mount the false drawer fronts to the drawers. Use playing cards or shims to help align the drawers to create the desired reveals. From the inside of the drawer, permanently attach the false drawer fronts to the drawers. You can now reinstall the drawer pulls.

Step 17

Finish the Miter Saw Station

Do any desired edge treatments and sanding before applying your finish of choice to the miter saw station. This is shop furniture and doesn't require a finish, but doing so will keep it looking nicer for longer.

Congratulations! You've Completed the Miter Saw Station Build!

Post your builds on Instagram and be sure to tag me @scottydwalsh
Be sure to include photos of the final product as well as photos from your build journey.

For more build plans, visit the link below:

www.scottwalsh.co